Responsibly Deploy AI in Quantitative Social Research

As a part of the Social Research Methods module

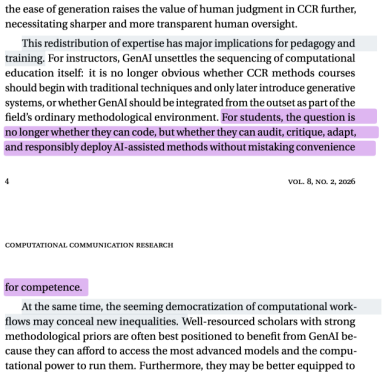
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Using causal and computational methods to evaluate
societal and political impacts of digital information technologies

November 6, 2026; Online

The goal of this session

... lower barrier (of computational skills)
... redistribution of expertise ...
"For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience..." (Zhou et al., 2026)



the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

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COMPUTATIONAL COMMUNICATION RESEARCH

for competence.

At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

Source: Zhou et al. (2026)

Introduction

1.1 AI is everywhere



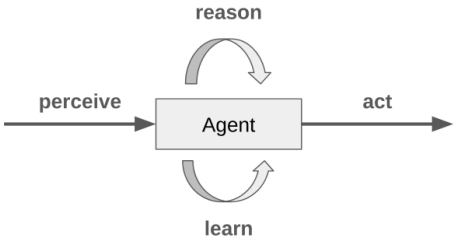
"Make it AI-powered first; Source: Bilibili, Oct 10, 2026

1.2 What is artificial intelligence?

- Computational agents that act intelligently

  (Poole & Mackworth, 2023)

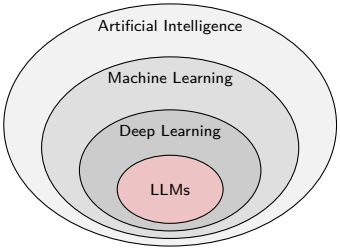
- But what is intelligence? See the figure



Source: Stanford CS221, Percy Liang

1.2 What is artificial intelligence?

- Computational agents that act intelligently (Poole & Mackworth, 2023);
- But what is intelligence? See the figure
- More importantly, please be aware of the differences ...



1.3 A simplified history of LLMs

- * Artificial intelligence is an old field developed decades ago (see Poole & Mackworth, 2023);
- * Pre-LLMs: early tools mainly classified, clustered, or predicted from human-produced traces (Zhou et al., 2026). [a]
- * Birth of LLMs: the Transformer (Vaswani et al., 2017), a model architecture that parallelizes training, enabling large datasets and model sizes
- Post-LLMs: we focus on the LLM era after 2020 (marked by GPT-3; Xu et al., 2026) and its implications for computational social science research

[a] Likewise, Percy Liang uses 2016, 2018, and today to divide three time periods; see https://cs336.stanford.edu/lectures/?trace=lecture_01

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

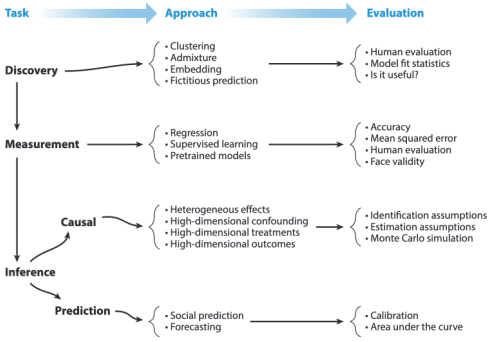
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

Excerpt from Vaswani et al. (2017)

Pre-LLMs AI

2.1 The use of AI in social science before LLMs

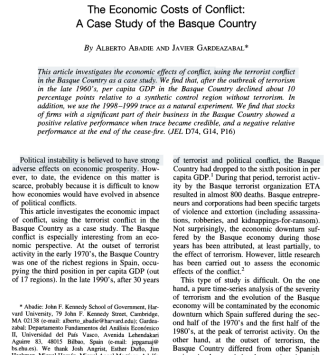
- Social science research often leverages machine learning for various tasks, according to the typology in Grimmer et al. (2021):
  - Discovery; e.g., text classification (Tutorial I)
  - Measurement; e.g., regression
  - Causal Inference; e.g., heterogeneous treatment effects
  - Prediction; e.g., forecasting social trends
- Take FIN528 Machine Learning for Finance / Advanced Machine Learning at PHBS if interested



Source: Grimmer et al. (2021) (p.23)

2.2 Example in Econ: Synthetic Control

- Abadie and Gardeazabal (2003) uses synthetic control units to estimate the economic costs of conflict in the Basque Country
- The control units are machine-constructed synthetic; similarly in Abadie et al. (2015)



The Economic Costs of Conflict:
A Case Study of the Basque Country

By Alberto Abadie and Javier Gardeazabal*

This article investigates the economic effects of conflict, using the terrorist conflict in the Basque Country as a case study. We find that, after the outbreak of terrorism in the late 1960's, per capita GDP in the Basque Country declined about 10 percentage points relative to a synthetic control region without terrorism. In addition, we use the 1998–1999 truce as a natural experiment. We find that stocks of firms with a significant part of their business in the Basque Country showed a positive relative performance when truce became credible, and a negative relative performance at the end of the cease-fire. (JEL D74, G14, P16)

Political instability is believed to have strong adverse effects on economic prosperity. However, to date, the evidence on this matter is scarce, probably because it is difficult to know how economies would have evolved in absence of political conflicts.

This article investigates the economic impact of conflict, using the terrorist conflict in the Basque Country as a case study. The Basque conflict is especially interesting from an economic perspective. At the outset of terrorist activity in the early 1970's, the Basque Country was one of the richest regions in Spain, occupying the third position in per capita GDP (out of 17 regions). In the late 1990's, after 30 years of terrorist and political conflict, the Basque Country had dropped to the sixth position in per capita GDP.[1] During that period, terrorist activity by the Basque terrorist organization ETA resulted in almost 800 deaths. Basque entrepreneurs and corporations had been specific targets of violence and extortion (including assassinations, robberies, and kidnappings-for-ransom). Not surprisingly, the economic downturn suffered by the Basque economy during those years has been attributed, at least partially, to the effect of terrorism. However, little research has been carried out to assess the economic effects of the conflict.[2]

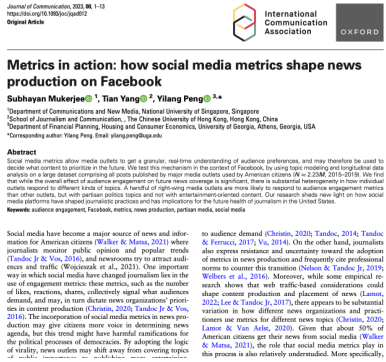
This type of study is difficult. On the one hand, a pure time-series analysis of the severity of terrorism and the evolution of the Basque economy will be contaminated by the economic downturn which Spain suffered during the second half of the 1970's and the first half of the 1980's, at the peak of terrorist activity. On the other hand, at the outset of terrorism, the Basque Country differed from other Spanish

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Source: Abadie & Gardeazabal (2003)

2.3 Example: Metrics in action

- Mukerjee et al. (2023) ask how audience engagement metrics shape news organizations' content production on Facebook
- Topic modeling (discovery) on all Facebook posts by major U.S. media outlets, combined with longitudinal analysis
- Mannual coding of topic categories



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Original Article

International Communication Association

OXFORD

Metrics in action: how social media metrics shape news production on Facebook

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Abstract

Social media metrics allow media outlets to get a granular, real-time understanding of audience preferences, and may therefore be used to decide what content to prioritize in the future. We test this mechanism in the context of Facebook, by using topic modeling and longitudinal data analysis on a large dataset comprising all posts published by major media outlets used by American citizens ($N = 2.22$M, 2015–2019). We find that while the overall effect of audience engagement on future news coverage is significant, there is substantial heterogeneity in how individual outlets respond to different kinds of topics. A handful of right-wing media outlets are more likely to respond to audience engagement metrics than other outlets, but with partisan politics topics and not with entertainment-oriented content. Our research sheds new light on how social media platforms have shaped journalistic practices and has implications for the future health of journalism in the United States.

Keywords: audience engagement, Facebook, metrics, news production, partisan media, social media

Social media have become a major source of news and information for American citizens (Walker & Matsa, 2021) where journalists monitor public opinion and popular trends (Tandoc Jr & Vos, 2016), and newsrooms try to attract audiences and traffic (Wojcieszak et al., 2021). One important way in which social media have changed journalism lies in the use of engagement metrics: these metrics, such as the number of likes, reactions, shares, collectively signal what audiences demand, and may, in turn dictate news organizations' priorities in content production (Christin, 2020; Tandoc Jr & Vos, 2016). The incorporation of social media metrics in news production may give citizens more voice in determining news agenda, but this trend might have harmful ramifications for the political processes of democracies. By adopting the logic of virality, news outlets may shift away from covering topics

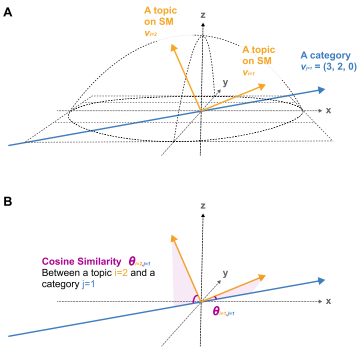
to audience demand (Christin, 2020; Tandoc, 2014; Tandoc & Ferrucci, 2017; Vu, 2014). On the other hand, journalists also express resistance and uncertainty toward the adoption of metrics in news production and frequently cite professional norms to counter this transition (Nelson & Tandoc Jr, 2019; Welbers et al., 2016). Moreover, while some empirical research shows that web traffic-based considerations could shape content production and placement of news (Lamot, 2022; Lee & Tandoc Jr, 2017), there appears to be substantial variation in how different news organizations and practitioners use metrics for different news topics (Christin, 2020; Lamot & Van Aelst, 2020). Given that about 50% of American citizens get their news from social media (Walker & Matsa, 2021), the role that social media metrics play in this process is also relatively understudied. More specifically,

Downloaded from https://academic.oup.com/joc/advance-article/doi/10.1093/joc/jqad012/7138729 by [illegible]

Source: Mukerjee et al. (2023)

2.4 Example: WIP

- We use semantic similarity score to measure the extent to which a topic on social media aligns with a specific topic category



Source: Work in progress

*TUT1

Sentiment classification using RoBERTa

Online material:
Google Colab Link

LLMs

3.1 Roles of LLMs in Computational Social Science

- Large language models (LLMs) are language models with roughly 1 billion or more parameters (Percy Liang; e.g., GPT, Claude)[1]
- A recent review article suggests four roles of LLMs in computational social science: subject, assistant, collaborator, and evaluator

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

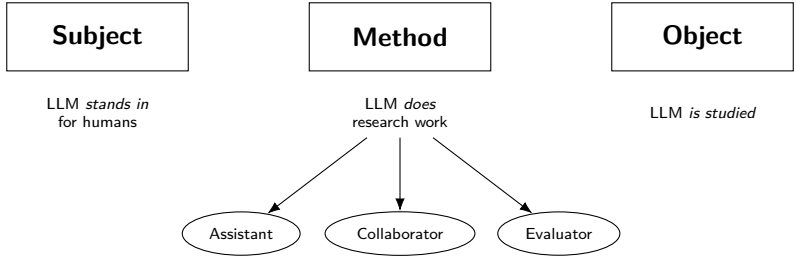
Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

Source: Xu et al. (2026)

[1] This is not a CS class, so it should be fine to use this definition.

3.1 Roles of LLMs in Computational Social Science

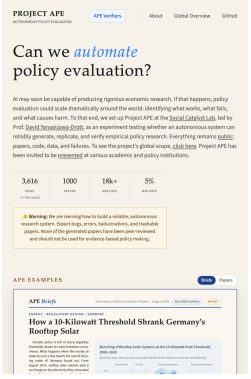
- Drawing on more perspectives, we propose a comprehensive categorization of LLM roles in CSS, elaborated in the following slides



Source: Four roles (subjects, assistants, collaborators, evaluators) categorized in Xu et al. (2026); four categories in H. Liang (2026); and two roles (a research tool, an object of inquiry) adapted from Zhou et al. (2026)

3.2 Example in Econ: Project APE

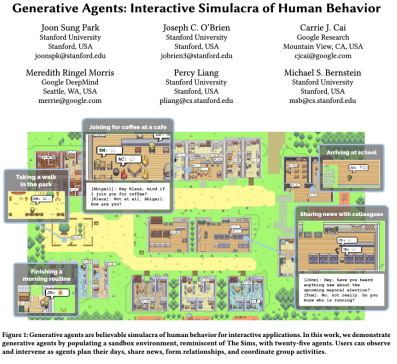
- "Project APE" (2026) uses large language models to automate policy evaluation



Source: "Project APE" (2026)

3.3 LLMs as Subjects: Categories

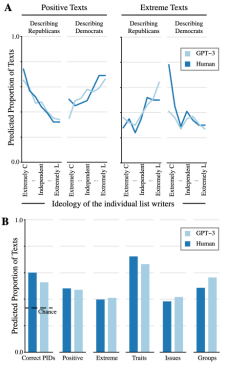
- LLMs serve as research subjects, according to Xu et al. (2026):
  - Generating survey responses (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - Simulating behavior sequences (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - Simulating social interactions (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



Generative agents in a simulated town; Park, O'Brien, et al. (2023)

3.4 LLMs as Subjects: Pros and Cons

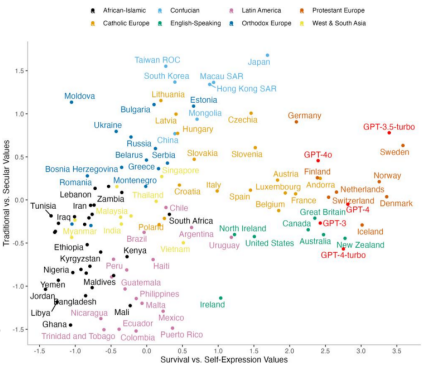
- **Pros:**
  - Successfully replicate human responses in surveys (Argyle, Busby, et al., 2023)
  - Produce believable individual and emergent social behaviors (Park, O'Brien, et al., 2023)
  - Replicate emergence of social conventions (Ashery et al., 2025)
  - etc.



Analysis of GPT-3 and human responses from the Lucid survey; Fig 3 in Argyle, Busby, et al. (2023)

3.4 LLMs as Subjects: Pros and Cons

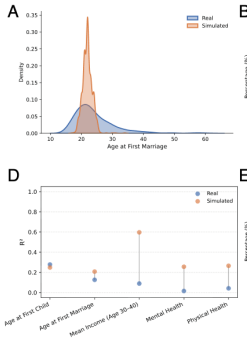
- **Cons:**
  - Poorly reflect "hard-to-reach populations" (Bollen et al., 2026); fail to capture experiential and cultural depth (Pataranutaporn et al., 2025)
  - Reflect certain partisan (Santurkar et al., 2023) or cultural biases (Tao et al., 2024)
  - Close to averages but with less variation (Bisbee et al., 2024); have strong, varying biases and have low variance in opinion or attitudinal research (Boelaert et al., 2025); likely to both misportray and flatten demographic representations (A. Wang et al., 2025); and tend to compress heterogeneity into simplified typological structures (Y. Xie & Xie, 2025)



LLM-generated answers on values deviate from those of different countries; Fig 1 in Tao et al. (2024)

3.4 LLMs as Subjects: Pros and Cons

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Failing to capture the dispersion of real-world distributions; Fig 3 in Y. Xie & Xie (2025)

3.5 LLMs as Subjects: Evaluation

- *Validity* and *reliability* remain concerns (Zhou et al., 2026); please refer to previous lectures by Prof. Ye



FIGURE 5-2
An Analogy to Validity and Reliability. A good measurement technique should be both valid (measuring what it is intended to measure) and reliable (yielding a given measurement dependably).
© Cengage Learning®

they are somewhat superior to those they study. For example, researchers often seek to uncover motivations that the social actors themselves are unaware of. You think you bought that new Turbo Tiger because of its high performance and good looks, but we know you're really trying to achieve a higher social status.

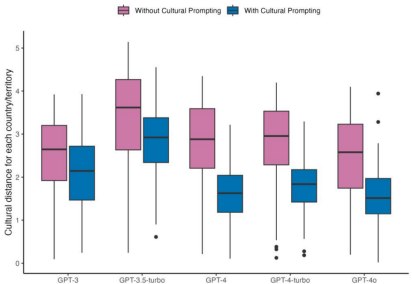
This implicit sense of superiority would fit comfortably with a totally positivistic approach (the biologist feels superior to the frog on the lab table), but it clashes with the more humanistic and typically qualitative approach taken by many social scientists. We'll explore this issue more deeply in Chapter 10. In seeking to understand they are escapes. The fact is that to commit oneself to a semiotic concept of culture and an interpretive approach to the study of it is to commit oneself to a view of ethnographic assertion as, to borrow W. B. Gallie's by now famous phrase, "essentially contestable." Anthropology, or at least interpretive anthropology, is a science whose progress is marked less by a perfection of consensus than by a refinement of debate. What gets better is the precision with which we vex each other. *(1973: 29)*

Ultimately, social researchers should look both to

Validity and reliability; from Babbie (2021)

3.6 LLMs as Subjects: Improvement

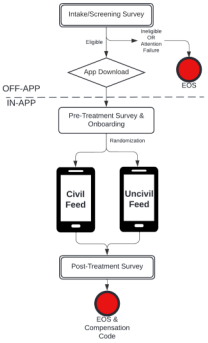
- *Validity* and *reliability* remain concerns (Zhou et al., 2026)
- Three strategies for improvement
  - Alignment (pre-use): Adjusting model outputs; mostly done by companies and in computer science studies
  - Optimization (in-use): **Fine-tuning or prompting** strategies to improve performance on specific tasks (e.g., Kolluri et al., 2025; Tao et al., 2024; Y. Xie & Xie, 2025)
  - Correction (post-use): Post-processing corrections; an emerging promise in the econometric and machine learning literature (e.g., Chernozhukov et al., 2018; Egami et al., 2024)



Prompting mitigates cultural biases; Source: Fig 2 in Tao et al. (2024)

3.7 LLMs as Methods: Assistant

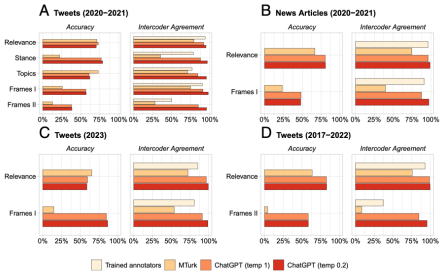
- Interviewing human participants (qualitative) (Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **Supporting experiments** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- Analysing textual content (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Synthetic communicative environment in a randomized experiment; Allamong et al. (2026)

3.7 LLMs as Methods: Assistant

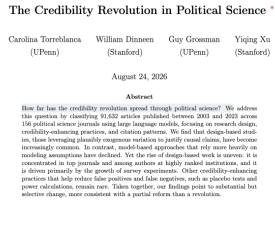
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- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Better annotation than crowd-workers; Gilardi et al. (2023)

3.8 Example in PS: Tracking Credibility Revolution

- Torreblanca et al. (2026) uses large language model to annotate papers in their review artical tracking the credibility revolution in political science; following Currie et al. (2020) and Goldsmith-Pinkham (2024) in economics.



The Credibility Revolution in Political Science *

Carolina Torreblanca (UPenn) William Dinneen (Stanford) Guy Grossman (UPenn) Yiqing Xu (Stanford)

August 24, 2026

Abstract

How far has the credibility revolution spread through political science? We address this question by classifying 91,632 articles published between 2003 and 2023 across 156 political science journals using large language models, focusing on research design, credibility-enhancing practices, and citation patterns. We find that design-based studies, those leveraging plausibly exogenous variation to justify causal claims, have become increasingly common. In contrast, model-based approaches that rely more heavily on modeling assumptions have declined. Yet the rise of design-based work is uneven: it is concentrated in top journals and among authors at highly ranked institutions, and it is driven primarily by the growth of survey experiments. Other credibility-enhancing practices that help reduce false positives and false negatives, such as placebo tests and power calculations, remain rare. Taken together, our findings point to substantial but selective change, more consistent with a partial reform than a revolution.

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Source: Torreblanca et al. (2026)

TUT2

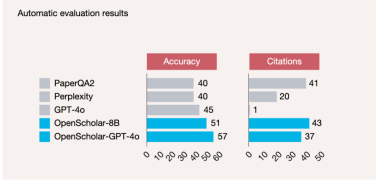
Topic classification using LLMs

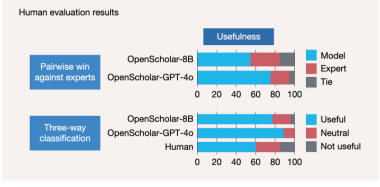
Don't forget about validation!

Online material: GitHub

3.9 LLMs as Methods: Collaborator

- **Reviewing literature** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- Proposing hypotheses (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)

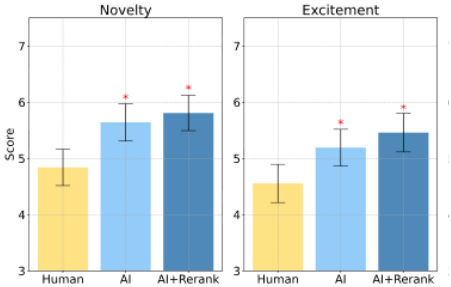




PhD-level literature review system powered by LLMs; see Asai et al. (2026)

3.9 LLMs as Methods: Collaborator

- Reviewing literature (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- **Proposing hypotheses** (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)



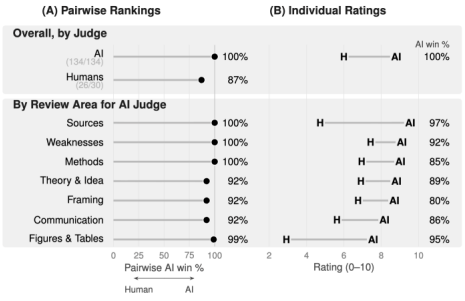
Greater novelty and excitement in LLM-generated hypotheses; see Si et al. (2024)

TUT3

Agentic Modelling in Data Analysis
Online material: GitHub

3.10 LLMs as Methods: Evaluator

- Evaluating research protocols (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **Critiquing research outcomes** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; W. Liang et al., 2024)



AI reviews dominate human reviews; source: (King & Worthington, 2026); not peer-reviewed but well designed

TUT4

Critically evaluate LLM-generated review comments
GitHub

3.11 LLMs as Objects

- "A communication technology that reconfigures production, circulation, interpretation, and power across media systems, raising critical questions about authenticity, access, and the reproduction of social inequalities." (Zhou et al., 2026)
- For example, Liu and Zhang (2026) evaluates how LLM-generated summaries shape public sentiment, where the treatment is the model itself.

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

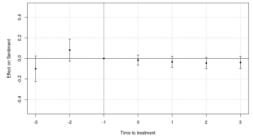


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

AIGS as objects in Liu & Zhang (2026)

Quiz

Propose one research question involving LLMs as objects in the online group chat

Please briefly clarify:

(a) the question of interest

(b) operationalization of your variables

(c) the most important figure to be presented

(d) expected contribution

3.12 *LLM for inference: New frontier

- Double/debiased machine learning enables valid causal inference while flexible ML models estimate nuisance functions

  (Chernozhukov et al., 2017)

- Building upon DML, Imai and Nakamura (2026) leverage the internal representations of generative AI to estimate causal effects of unstructured treatments such as texts and images



PNAS RESEARCH ARTICLE | SOCIAL SCIENCES

Leveraging generative AI for causal inference with unstructured data

Kosuke Imai[a,1,2] and Kentaro Nakamura[a,1]

Affiliations are included on p. 5.

Edited by David M. Blei, Columbia University, New York, NY; received October 24, 2025; accepted July 29, 2026 by Editorial Board Member Kenneth W. Wachter

We introduce GenAI-Powered Inference (GPI), a statistical framework for causal inference using unstructured data, including text and images. GPI leverages open-source pretrained Generative AI (GenAI) models—such as large language models and diffusion models—not only to generate unstructured data at scale but also to extract low-dimensional representations that are guaranteed to capture their underlying structure. Applying machine learning to these representations, GPI enables estimation of causal effects while quantifying estimation uncertainty. Unlike existing approaches to representation learning, GPI does not require fine-tuning of GenAI models, making it computationally efficient and broadly accessible. We illustrate the versatility of the GPI framework through three applications: 1) estimating the effects of Chinese social media censorship while adjusting for textual confounders, 2) isolating the impact of specific image features from that of other correlated features in the same image, and 3) assessing the persuasiveness of political rhetoric. An open-source software package is available for implementing GPI.

generative AI | causal inference | machine learning | unstructured data

The emergence of Generative AI (GenAI), including large language models and diffusion models, has transformed society and scientific research. In this paper, we show that GenAI can also serve as a powerful tool for causal inference with unstructured data such as text and images. Specifically, we introduce a statistical framework, GenAI-Powered Inference (GPI), which leverages open-source GenAI models to improve causal inference with high-dimensional unstructured data.

We illustrate the power of GPI through the following three applications:

1. **Text as confounder**: Estimating the causal effects of a treatment variable by adjusting for latent confounders embedded in textual data
2. **Image as Treatment**: Isolating the impact of specific image features from that of other correlated features in the same image
3. **Structural model of text**: Fitting a structural model that incorporates both observed and latent textual features as covariates

A common challenge in these settings is that while high-dimensional unstructured data are observed, we do not know, a priori, which features of such data serve as relevant confounders.

GPI addresses this challenge by leveraging the ability of GenAI to generate or reproduce unstructured data and extracting lower-dimensional vector representations that are guaranteed to contain all the relevant information. Machine learning is then applied to these representations to estimate a *deconfounder*, which is a summary of confounding

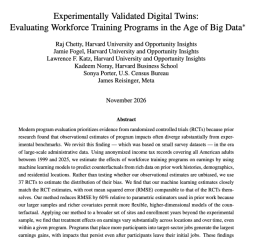
Significance

We introduce GenAI-Powered Inference (GPI), a unified methodological framework that integrates open-source large language and diffusion models into causal inference. GPI uses generative AI (GenAI) models to extract low-dimensional representations that retain the information necessary to generate complex unstructured data. This enables researchers to estimate causal effects without directly modeling high-dimensional objects or fine-tuning foundation models. We show that estimating and adjusting for a deconfounder leads to valid causal inference with unstructured data. By leveraging pretrained GenAI models while avoiding restrictive modeling assumptions, GPI delivers strong empirical performance across diverse settings. We demonstrate the versatility of the framework in three distinct applications and provide open-source software to facilitate its use in applied research.

Source: Imai & Nakamura (2026)

3.12 *LLM for inference: New frontier

- Double/debiased machine learning enables valid causal inference while flexible ML models estimate nuisance functions
  (Chernozhukov et al., 2017)
- Building upon DML, Imai and Nakamura (2026) leverage the internal representations of generative AI to estimate causal effects of unstructured treatments such as texts and images
- DML estimators, moreover, are (a) closed to RCT baselines yet have slightly (b) lower RMSE (Chetty et al., 2026)



Experimentally Validated Digital Twins:
Evaluating Workforce Training Programs in the Age of Big Data*

Raj Chetty, Harvard University and Opportunity Insights
Jamie Fogel, Harvard University and Opportunity Insights
Lawrence F. Katz, Harvard University and Opportunity Insights
Kadeem Noray, Harvard Business School
Sonya Porter, U.S. Census Bureau
James Reisinger, Meta

November 2026

Abstract

Modern program evaluation prioritizes evidence from randomized controlled trials (RCTs) because prior research found that observational estimates of program impacts often diverge substantially from experimental benchmarks. We revisit this finding — which was based on small survey datasets — in the era of large-scale administrative data. Using anonymized income tax records covering all American adults between 1999 and 2025, we estimate the effects of workforce training programs on earnings by using machine learning models to predict counterfactuals from rich data on prior work histories, demographics, and residential locations. Rather than testing whether our observational estimates are unbiased, we use 37 RCTs to estimate the distribution of their bias. We find that our machine learning estimates closely match the RCT estimates, with root mean squared error (RMSE) comparable to that of the RCTs themselves. Our method reduces RMSE by 68% relative to parametric estimators used in prior work because our larger samples and richer covariates permit more flexible, higher-dimensional models of the counterfactual. Applying our method to a broader set of sites and enrollment years beyond the experimental sample, we find that treatment effects on earnings vary substantially across locations and over time, even within a given program. Programs that place more participants into target-sector jobs generate the largest earnings gains, with impacts that persist even after participants leave their initial jobs. These findings provide guidance for improving job training programs and, more broadly, illustrate a scalable approach to program evaluation that uses observational data to construct experimentally validated "digital twins."

* Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has reviewed appropriate access and use of confidential data and has reviewed the results for disclosure avoidance protection (Project 7519874 - CBDRB-FY22-CBSM14-046, CBDRB-FY23-CBSM14-039, CBDRB-FY23-CES027-004, CBDRB-FY23-CES027-006, CBDRB-FY23-CES027-010, CBDRB-FY24-CES004-005, CBDRB-FY24-CBSM14-003, CBDRB-FY24-CES027-002, CBDRB-FY25-CES027-001, CBDRB-FY25-CES027-003, CBDRB-FY26-CES027-004, CBDRB-FY26-CES027-006, CBDRB-FY26-CES027-007). Publication of this research was supported by the Chan Zuckerberg Initiative, Meta, Schmidt Sciences, Stripe, Inc., Arnold Ventures, Overdeck Family Foundation, and Spotify USA Inc. through their support of the Digital Economy Project at Harvard Kennedy School. Raj Chetty leads Opportunity Capital, an organization that builds on the methods developed in this paper to invest in training programs. We are very grateful to Rohit Baichwal, Corey Gelb-Richard, Jahui Jie, Josephine Kim, Samir Rois, and other Opportunity Insights pre-doctoral fellows for outstanding research assistance. We used AI to edit and proofread this manuscript. We thank Nathan Hendren, Guido Imbens, Yechan Park, Jonathan Roth, Jesse Shapiro, Davide Viviano, Joe Wardelmann, and Yuqing Xu for helpful comments and suggestions. We are grateful to Ben Glass, Jeremy Hildreth, Jonathan Latner, Ryan Perez, and Jonathan Rothbaum at Economic Mobility Corporation for providing data from randomized evaluations and answering questions about the programs and their implementation. We also thank Tyse Up United, Per Scholas, and Project QUEST for providing additional data, and the Social Finance Institute for helping facilitate data access. This research was funded by the Andrew Carnegie Foundation, Chan Zuckerberg Initiative, Gates Foundation, Overdeck Family Foundation, Sanford Philanthropies, and the Walton Family Foundation.

Source: Chetty et al. (2026)

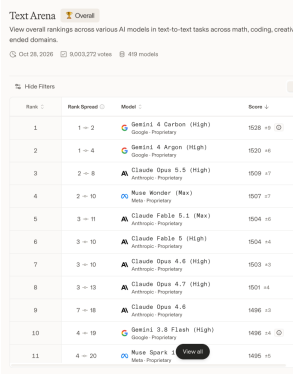
Deploy Responsively

4.1 LLMs deployed responsibly: Selection

- Consider a model as a function, $f(data, algo, objective)$; differences in each part substantially shape its in-domain capabilities (like better coding or writing)
- In this sense, we may need to choose different models for different tasks, depending on their strengths and limitations

4.1 LLMs deployed responsibly: Selection

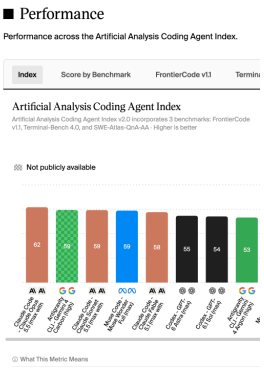
- For example, you may expect Gemini Carbon to excel at writing tasks, whereas Claude Opus 5.5 outperforms in engineering



Source: https://arena.ai/leaderboard/chat/text (Nov 3, 2026)

4.1 LLMs deployed responsibly: Selection

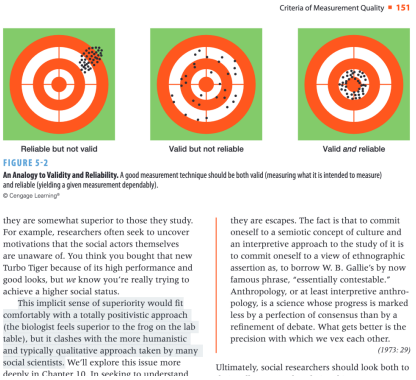
- For example, you may expect Gemini Carbon to excel at writing tasks, whereas Claude Opus 5.5 outperforms in engineering
- In general, model labs follow different commercialization strategies; a lab whose business model targets engineering tasks may prioritize coding capabilities (e.g., Anthropic); please check benchmarks / review the literature / do a pilot test before using a model



Source: https://artificialanalysis.ai/agents/coding-agents#coding-agents-index (Nov 3, 2026)

4.2 LLMs deployed responsibly: Validity and Reliability

- Again, always verify the validity and reliability of the model outputs before using them in your research



Criteria of Measurement Quality ■ 151

Reliable but not valid | Valid but not reliable | Valid *and* reliable

FIGURE 5-2

An Analogy to Validity and Reliability. A good measurement technique should be both valid (measuring what it is intended to measure) and reliable (yielding a given measurement dependably).

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they are somewhat superior to those they study. For example, researchers often seek to uncover motivations that the social actors themselves are unaware of. You think you bought that new Turbo Tiger because of its high performance and good looks, but *we* know you're really trying to achieve a higher social status.

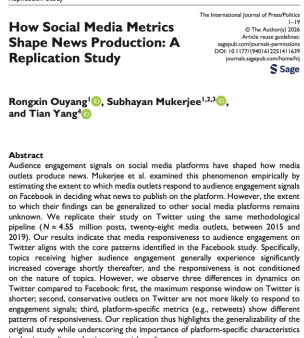
This implicit sense of superiority would fit comfortably with a totally positivistic approach (the biologist feels superior to the frog on the lab table), but it clashes with the more humanistic and typically qualitative approach taken by many social scientists. We'll explore this issue more deeply in Chapter 10. In seeking to understand they are escapes. The fact is that to commit oneself to a semiotic concept of culture and an interpretive approach to the study of it is to commit oneself to a view of ethnographic assertion as, to borrow W. B. Gallie's by now famous phrase, "essentially contestable." Anthropology, or at least interpretive anthropology, is a science whose progress is marked less by a perfection of consensus than by a refinement of debate. What gets better is the precision with which we vex each other. *(1973: 29)*

Ultimately, social researchers should look both to

Validity and reliability; from Babbie (2021)

4.3 LLMs deployed responsibly: Reproducibility

- Reproducibility is important to science
- Ensure that the AI part of your research is reproducible by:
  - Preferring API-based access over chat-based access
  - Documenting model versions, prompts, thinking level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the random seed and temperature (search for details)



Replication Study

The International Journal of Press/Politics
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How Social Media Metrics Shape News Production: A Replication Study

Rongxin Ouyang[1], Subhayan Mukerjee[1,2,3], and Tian Yang[4]

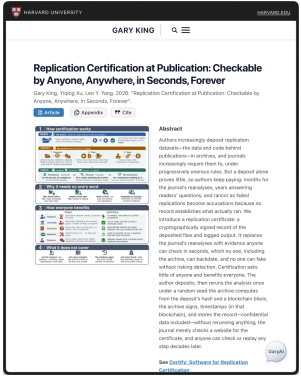
Abstract

Audience engagement signals on social media platforms have shaped how media outlets produce news. Mukerjee et al. examined this phenomenon empirically by estimating the extent to which media outlets respond to audience engagement signals on Facebook in deciding what news to publish on the platform. However, the extent to which their findings can be generalized to other social media platforms remains unknown. We replicate their study on Twitter using the same methodological pipeline ($N \approx 4.55$ million posts, twenty-eight media outlets, between 2015 and 2019). Our results indicate that media responsiveness to audience engagement on Twitter aligns with the core patterns identified in the Facebook study. Specifically, topics receiving higher audience engagement generally experience significantly increased coverage shortly thereafter, and the responsiveness is not conditioned on the nature of topics. However, we observe three differences in dynamics on Twitter compared to Facebook: first, the maximum response window on Twitter is shorter; second, conservative outlets on Twitter are not more likely to respond to engagement signals; third, platform-specific metrics (e.g., retweets) show different patterns of responsiveness. Our replication thus highlights the generalizability of the original study while underscoring the importance of platform-specific characteristics in shaping media production on social media.

Replicating Mukerjee et al. (2023) on Twitter; Source: Ouyang et al. (2026)

4.3 LLMs deployed responsibly: Reproducibility

- Reproducibility is important to science
- Ensure that the AI part of your research is reproducible by:
  - Preferring API-based access over chat-based access
  - Documenting model versions, prompts, thinking level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the random seed and temperature (search for details)



Replication certification at publication; Source: King et al. (2026)

*TUT5 & 6

Multimodal object detection and controlled stimulus generation
Online material: GitHub

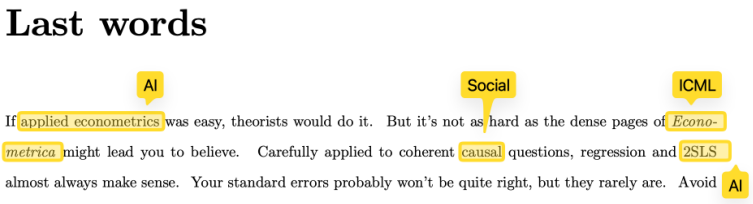
Takeaways

5.1 Takeaways

- AI models (including LLMs) are just tools for high-dimensional data analysis; they can be subjects, objects, and methods in quantitative social research
- As methods, they are expected to be used in a responsible way in which validity, reliability, and reproducibility are carefully considered
- Like any other methods, although they evolve rapidly, their use should rest on our taste for good questions in research, industry, and life, which remains central to higher education.

5.1 Takeaways

- "... Be skeptic... don't panic!" (Angrist & Pischke, 2009)



Source: Angrist & Pischke (2009)

Resources

6.1 Resources

- **Locate the most appropriate AI model for your task**
  - Artificial Analysis https://artificialanalysis.ai/
  - Arena.ai, blind benchmarking for AI models https://arena.ai/leaderboard/
- **Find problems in your manuscript**
  - Refine, publication-level technical review; adopted by AEA https://www.refine.ink/
  - OpenAIReview, open-source multi-agent manuscript review solution by UChicago HAI https://github.com/ChicagoHAI/OpenAIReview
- **Double-check references and citations**
  - CiteVerify, "a local tool for verifying references in research papers", https://github.com/cuihuash/CiteVerify
  - Chatting with frontier models also helps

6.1 Resources

- **Organize multiple AI models and services**
  - For macOS, magpie, a local AI router manager for Mac https://github.com/yetone/magpie
  - For other platforms, cc-switch, https://github.com/farion1231/cc-switch
- **Skills**
  - AER-skills, https://github.com/brycewang-stanford/AER-skills
  - Claude Code setup by Pedro H. C. Sant'Anna, https://psantanna.com/claude-code-my-workflow/
- **Read scientific literature in your own language**
  - PDFMathTranslate, read scientific PDFs in your language https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)
- **More examples if you find them helpful...**
  - 打造 AI 科学家: DeepSeek 与 ChatGPT 应用实战手册 (机械工业出版社, 2026)

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